

SAS821S Capstone Project — Group Contribution Statement

Topic T18 · Insurance Claims Fraud, Identity Abuse and Data-Leakage Analytics

1. Group details

Member A (Data and Modelling Lead): Shanon Chiwara, Student No. 223127744

Member B (Security Engineering and Intelligence Lead): Ephath Shimhanda, Student No. 216069076

2. Purpose of this statement

This statement records how the two group members divided, executed and jointly reviewed the work for the SAS821S capstone project, per Section 4 (Group structure and member roles) and the RACI allocation defined in the Project Charter, Section 9. It supports the marking criteria requiring documented, balanced individual contribution.

3. Workstream contribution

Workstream / deliverable	Primary owner	Reviewed by
Data acquisition, cleaning, feature engineering, data dictionary (02_data/, generate_document_text_source.py, generate_portal_auth_device_data.py, link_and_engineer_sources.py)	Member A	Member B
Exploratory analysis and visual baselines (run_eda_baseline.py)	Member A	Member B
Supervised fraud model and unsupervised/anomaly model, evaluation (supervised_fraud_model.py, unsupervised_anomaly_model.py)	Member A	Member B (security validation)
Predictive risk / adversarial and robustness testing (adversarial_robustness_testing.py)	Member A	Member B
Security investigation, incident timeline, access findings (incident_timeline_investigation.py)	Member B	Member A
Security intelligence requirements and enrichment products	Member B	Member A
Control-scenario simulation (security_control_simulation.py)	Member B (accountable)	Member A (analytics support)
Text mining / NLP workflow (nlp_text_mining.py)	Member B	Member A
Analyst-facing prototype / dashboard (build_dashboard.py)	Shared	Shared
Final documentation, README, reproducibility bundle, presentation	Shared	Shared

3.1 Note on version-control history

The group used GitHub Desktop for version control. Both members wrote and edited code throughout the project, but commits to the shared repository were pushed from one member's (Shanon Chiwara's) GitHub Desktop account, so the commit log shows a single committer rather than two. This reflects how commits were pushed, not who wrote the work: code, notebooks and documentation were developed and reviewed jointly by both members before being committed, consistent with the workstream ownership recorded in Section 3 above.

Both members confirm that:

- Each member implemented at least one analytical method independently and reviewed at least one component owned by the partner.
- Both members made meaningful contributions to code, analysis, writing and presentation preparation.
- Both members understand the complete end-to-end solution, not only the components they were primarily responsible for.
- Version-control history, meeting notes and issue tracking were retained as supporting contribution evidence.

4. Declaration

We declare that the contribution split described above is an accurate and honest record of how this project was completed, and that both of us are able to explain and defend every component of the submitted solution during the oral question-and-answer session.

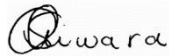
5. Signatures

Name: Shanon Chiwara

Student number: 223127744

Role: Member A — Data and Modelling Lead

Signature:

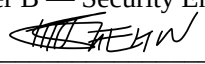


Date: 24/09/26

Name: Ephath Shimhanda

Student number: 216069076

Role: Member B — Security Engineering and Intelligence Lead

Signature: 

Date: _____25/09/2026_____